

APAR POKHREL

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Education

University of Texas at Arlington

August 2024 – May 2026

M.S. in Computer Science

GPA: 4.0/4.0

- **Advisor:** Diegeo Patino
- **Thesis:** pending

University of Texas at Arlington

May 2022

B.S. in Computer Science, Minor in Mathematics

GPA: 3.81/4.0

Research Interests

- Computer Vision & 3-D Reconstruction
- Neural Networks & Deep Learning
- Neural Rendering & Generative Models
- Computational Biology & Medical Imaging
- LLMs & Vision LLMs
- Human/Robot Interaction

Relevant Coursework

- Neural Networks
- Data Mining
- Algorithm Analysis
- Artificial Intelligence
- Machine Learning
- Computer Vision
- Bioinformatics
- Computer Networks

Research Experience

Research Assistant

May 2025 – Present

PRIMaL Lab

Arlington, Texas

- Working on diffusion models and point clouds for 3-D scene understanding and reconstruction.
- Collaborating with team members and working towards my thesis

Work Experience

Software Engineer

September 2022 – February 2024

SAE International

Warrendale, Pennsylvania

- **eAuditNet** | Java, JavaScript, Oracle, Spring MVC + JSP
 - * Resolved bug tickets by examining Kibana logs and engaging clients, reducing repeat incidents by 25%.
 - * Developed store procedures to optimize audit and expense processes, slashing support ticket response time by 30% and assisted in resolving data exchange failures by dissecting failure codes and server logs.
 - * Successfully designed and integrated email configurations to streamline audit scheduling, invoices, and payments.
- **WMI/WMC** | Java, ReactJS, PostgreSQL
 - * Improved global WMI code allocation logic for car manufacturers by 70% using indexing & caching table lookups.
 - * Created schema designs, identified data mappings, and transformation requirements for PostgreSQL migration.
 - * Created dynamic reports across global code confirmation, cancellation, and verification areas, streamlined report delivery via SendGrid API, resulting in a 60% reduction in report distribution time.

Student Developer, IEEE Robotics

August 2021 – May 2022

The University of Texas at Arlington

Arlington, Texas

- Collaborated on build for an underwater ROV capable of performing dive, submerge, and travel maneuvers.
- Developed an interactive ReactJS control interface with integrated live video streaming for real-time maneuverability.

Teaching Experience

Graduate Teaching Assistant

August 2024 – Present

University of Texas at Arlington

Arlington, Texas

- Support faculty in instruction of Object Oriented Programming for 120+ students, enhance student performance through review sessions, proctor exams, grade assignments, and streamline grade entries via **auto-grader**.

Drone Design

Summer 2025

Upward Bound Math & Science Center

Arlington, Texas

- Develop and lead an intensive project-based curriculum introducing students to drone aerodynamics, control systems, electronics, and hands-on design using simulation and prototyping tools.

Bioinformatics & Medical Research

Summer 2025

Upward Bound Math & Science Center

Arlington, Texas

- Design and deliver bioinformatics module on sequence alignment tools for genomic analysis, protein structures, and introduce machine learning models for biological data classification, clustering, and medical image segmentation.

BugHouse Tutor

University of Texas at Arlington

August 2024 – Present

Arlington, Texas

- Assist Computer Science & Engineering students across lower and upper-level CSE coursework.

Peer Academic Leader

University of Texas at Arlington

January 2021 – May 2022

Arlington, Texas

- Guided first-year engineering students on academic affairs policy, major exploration, and engineering practices.

Projects

Optimizing YOLOv8 for Parking Space Detection – PKLot | *Python, PyTorch, CUDA*

May 2025

- Achieved 0.986 mAP@0.5:0.95 score by customizing YOLOv8 with optimized backbones (ResNet-18, VGG16, EfficientNetV2, Ghost-P2) on the PKLot dataset.
- **[Pre-Print]** Conducted a detailed study of backbone architectures, analyzing precision–recall trade-offs, inference latency, and computational efficiency for real-time performance.

Hackathons | *Flask, ReactJS, Bootstrap, Axios, Python, Tensorflow, Google Collab*

May 2022 – September 2022

- **Insure It:** Mobile app that parses consumer receipts for easy insurance quote suggestions. (1st place HackUTA)
- **CareHive:** ML model that detects neonatal pneumonia from chest radiographs. (3rd place @Hackdemonium 2.0)
- **AgroHive:** ML model that predicts 38 crop diseases among 13 crop varieties to improve crop yield. (HackTX)
- **10 years with Jake:** Interactive game that stimulates a 10-year State Farm home insurance plan. (HackDFW)

Honors & Awards

- **Maverick Academic Scholarship.** Annual \$24,000 scholarship. 2018 - 2022
- **Graduate Teaching Assistantship,** Annual \$14,000 waiver. 2024
- **Department of Education Dean's Scholarship.** 2024

Talks & Presentations

- **Medical Image Segmentation using CNN & Transformers.** Upward Bound Math & Science Center. 2025
- **Vision-Based Drone Navigation: Deep Learning Approaches for Object Detection and Path Planning.** Upward Bound Math & Science Center. 2025
- **Optimizing YOLOv8 for Parking Space Detection: Comparative Analysis of Custom Backbone Architectures.** University of Texas at Arlington. 2024
- **From Photos2Anime: Exploring Generative Adversarial Networks.** University of Texas at Arlington. 2022

Technical Skills

Languages: Python, JavaScript, Java, SQL, C/C++

Database: Oracle, MySQL, PostgreSQL, MongoDB, Neo4j

APIs & Integration: OpenAPI, FastAPI, Swagger, REST

Libraries & Frameworks : $\text{\LaTeX}$, ReactJS, Axios, Jest, NodeJS, ExpressJS, Bootstrap, TailwindCSS

Cloud & Developer Tools: AWS, GCP, Git, Gitlab CI/CD, Jupyter, Jira, Postman, Elasticsearch

Machine Learning: PyTorch, NumPy, Pandas, scikit-learn, NLTK